LABWORK 2

*This lab aims to teach **how to call a function inside another function** and how to **transfer data between functions**.*

Task 1: Cone Volume and Surface Area Calculator (30 pts)

> Learn to use functions by calculating the volume and surface area of a cone.

Function 1:

- **Name of the function:** `calculate_circle_area`
- **Return type:** `double`
- **Input parameters:** 1 (`double radius`)
- **Description:** Calculates and returns the area of a circle given its radius. (Use 3.14159 for π.)

Function 2:

- **Name of the function:** `calculate_slant_height`
- **Return type:** `double`
- **Input parameters:** 2 (`double radius`, `double height`)
- **Description:** Calculates and returns the slant height (*l*) of a cone using the Pythagorean theorem. (*l = sqrt(radius² + height²)*)

Function 3:

- **Name of the function:** `calculate_cone_volume`
- **Return type:** `double`
- **Input parameters:** 2 (`double radius`, `double height`)
- **Description:** Calculates and returns the volume of a cone. This function must call `calculate_circle_area` to get the base area. *(Volume = (1/3) × Base Area × Height)*

Function 4:

- **Name of the function:** `calculate_cone_surface_area`
- **Return type:** `double`
- **Input parameters:** 2 (`double radius`, `double height`)
- **Description:** Calculates and returns the total surface area of a cone. This function must call `calculate_circle_area` to get the base area and `calculate_slant_height` to find the slant height. *(Total Surface Area = Base Area + π × radius × slant height)*

Main Function:

- **Name of the function:** `main`

- **Return type:** `int`

- **Input parameters:** none

- **Description:**
    1. Request two inputs from the user: one for the cone's radius and the other for its height (both as doubles).
    2. Call the `calculate_cone_volume` function to determine the cone's volume.
    3. Call the `calculate_cone_surface_area` function to determine the cone's total surface area.
    4. Print the calculated volume and surface area to the screen.

Task 2: Water Bill Calculator (70 pts)

> Learn to use functions within other functions by calculating a water bill based on age, consumption, and house type.

Scenario

A user's **monthly water bill** is calculated according to:

- **House Type:** Apartment (A) or Villa (V)
- **Age:** (for discount)
- **Monthly Consumption (m^3)**

The final bill depends on:

1. The base rate (per m^3)
2. The total consumption
3. Discount or surcharge depending on the user's age

Function 1

- **Name:** `getBaseRate`
- **Return Type:** `double`
- **Input Parameters:** `char houseType`
- **Description:** Returns the base price per cubic meter (m^3):
    - Apartment (A/a): **5.0 TL/m^3**
    - Villa (V/v): **6.5 TL/m^3**

Function 2

- **Name:** `calculateConsumptionCost`
- **Return Type:** `double`
- **Input Parameters:** `double baseRate, double cubicMeters`
- **Description:** Calculates the cost before discounts or extra charges: `cost = baseRate * cubicMeters`

Function 3

- **Name:** `applyAgeAdjustment`
- **Return Type:** `double`
- **Input Parameters:** `int age, double cost`
- **Description:**
  - If age < 18 → 10% discount
  - If 18 ≤ age < 65 → no change
  - If age ≥ 65 → 15% discount Returns the adjusted cost.

Function 4

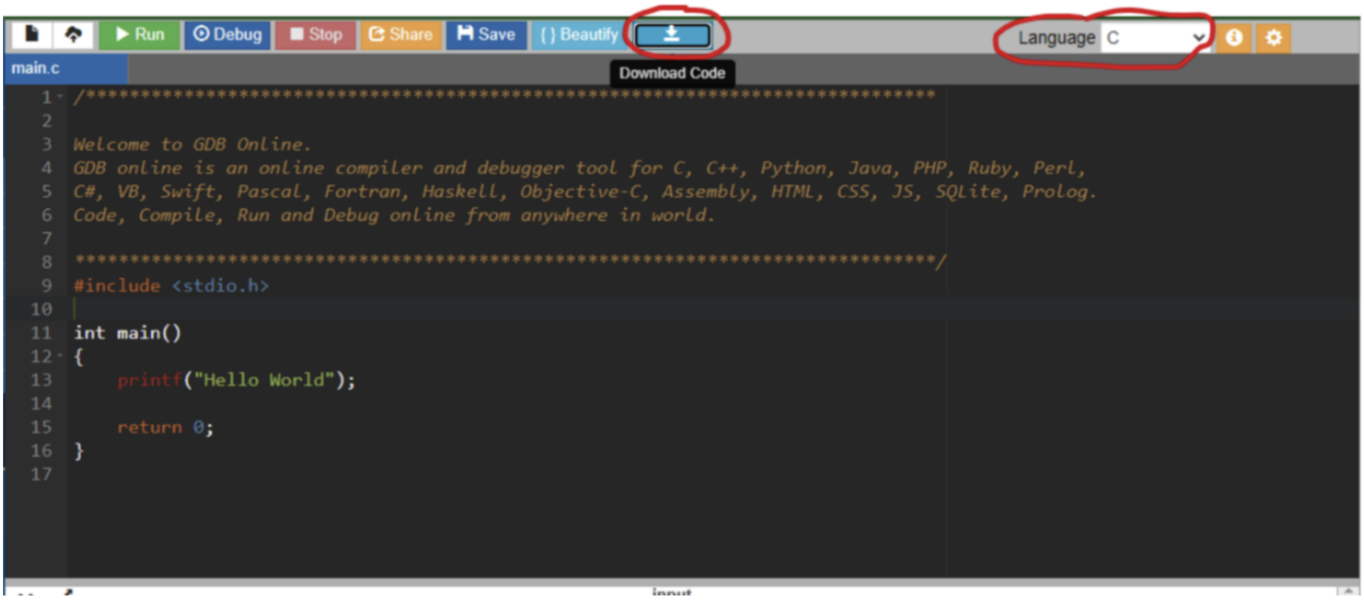
- **Name:** `calculateTotalBill`
- **Return Type:** `double`
- **Input Parameters:** `int age, char houseType, double cubicMeters`
- **Description:** This function **calls all other functions**:
  1. Calls `getBaseRate` to determine the unit price
  2. Calls `calculateConsumptionCost` to find base cost
  3. Calls `applyAgeAdjustment` to find final bill Returns the total amount.

Main Function

- **Name:** `main`
- **Return Type:** `int`
- **Description:**
  - Requests user input: `age`, `houseType`, and `monthly water consumption (m³)`
  - Calls `calculateTotalBill`
  - Prints the total water bill

Restrictions

- If you are not sure something is free to use or not, please ask your assistant.
- You have to complete the labwork using the functions provided. If you complete the labwork without using functions, your work will not be graded.
- Mobile phone and internet usage are not allowed.
- You can only access yulearn and the online C compiler https://www.onlinegdb.com/
- Do not forget to select the language if you use onlinegdb. When you finish your work, you can download your code by using the download code button on top of the window.



Submission

- Submit your C file with the format "name_surname.c" (use your name and surname).
- Do not submit a Word document, text file, or executable (a.out).